Electrochemical Device Modelling

End-to-End Computational Model of the Multiplexed Wound Protease Cartridge

Eton iGEM 2026 — Dry Lab

MMP-9 / HNE Ratiometric SWV Sensor

[title=Scope and motivation] There is no physical prototype of the electrochemical cartridge. Every physical process from wound-fluid biology through to the clinical readout number must be modelled mathematically and computationally. The chain is fully coupled: outputs of each layer feed as inputs to the next. The model must produce falsifiable, quantitative predictions (predicted signal F_k vs known protease concentration) that can be validated against wet-lab calibration data, satisfying the iGEM gold criterion that models predict specific experimental outcomes that wet lab then confirms.
This document enumerates every sub-component that requires explicit mathematical treatment across 16 modelling layers, with the governing equations, key parameters, and inter-layer coupling for each.

Layer 0: Wound biology — system input

The model input is the active concentration of MMP-9 and HNE in wound fluid at the time of sampling. These are not directly observable; they must be derived from the wound biochemical ODE system (Pillar 2, `report_project_proposal.tex`).

ODE system for protease activation

$$\frac{d[\text{proMMP-9}]}{dt} = -k_{\text{act}}[\text{plasmin}][\text{proMMP-9}] + k_{\text{syn}} \tag{1}$$

$$\frac{d[\text{MMP-9}]}{dt} = k_{\text{act}}[\text{plasmin}][\text{proMMP-9}] - k_{\text{TIMP}}[\text{TIMP-1}][\text{MMP-9}] \tag{2}$$

$$\frac{d[\text{HNE}]}{dt} = k_{\text{rel}} - k_{\text{A1AT}}[\text{A1AT}][\text{HNE}] \tag{3}$$

Outputs: $[\text{MMP-9}]_{\text{active}}$ and $[\text{HNE}]_{\text{active}}$ as functions of wound state and time since debridement.

Environmental parameters

- **pH**: range 6.5–8.2 (chronic to acute). Affects enzyme k_{cat} via pH-activity profile, formal potential E°_{MB} (shifts $-59\,\text{mV/pH}$ for the $2e^-/2H^+$ MB couple), and Debye screening.
- **Ionic strength** I: physiological $\approx 150\,\text{mM}$. Debye length $\lambda_D = \sqrt{\varepsilon\varepsilon_0 RT/(2F^2 I)} \approx 0.8\,\text{nm}$.
- **Temperature**: operating range 20–37°C. Every rate constant downstream is temperature-dependent.
- **Oxidative species**: H_2O_2, hypochlorite can oxidise the Cys–Au thiol bond. Model as first-order SAM degradation: $d\Gamma/dt|_{\text{ox}} = -k_{\text{ox}}[\text{ROS}]\Gamma$.
- **Matrix proteins**: albumin, fibrinogen contribute to non-specific adsorption (NSA) on gold. NSA fraction θ_{NSA} blocks probe-electrode area and increases baseline capacitance.
- **Interfering proteases**: cathepsins B/D/G, plasmin, urokinase. Each contributes off-target cleavage of the probe substrates with its own $k_{\text{cat,j}}$ and $K_{m,j}$ (from MEROPS). Must be included as additive cleavage terms.

Layer 1: Sample collection and pre-analytical

Swab absorption and elution

Let V_{abs} = volume absorbed by swab (20–200 μL), η_{elut} = elution efficiency (0.4–0.8 for proteins, size-dependent).

Concentration in eluate: $[E]_{\text{eluate}} = [E]_{\text{wound}} \times \eta_{\text{elut}}$.

Analyte stability during transit

First-order degradation from collection to assay (time t_{transit}):

$$[E]_{\text{eluate}}(t_{\text{transit}}) = [E]_{\text{eluate}}(0) \times \exp(-k_{\text{deg}}\, t_{\text{transit}}) \tag{4}$$

k_{deg} from literature self-digestion rates or measured stability assay.

Net dilution into cartridge

$$[E]_{\text{chamber}} = [E]_{\text{wound}} \times \eta_{\text{elut}} \times \exp(-k_{\text{deg}} t_{\text{transit}}) \times D_f \tag{5}$$

where $D_f = V_{\text{eluate}}/V_{\text{chamber}}$ is the dilution factor.

Layer 2: Cartridge inlet and capillary fill

Chamber geometry

Chamber dimensions: $20 \times 5\,\text{mm}$ footprint, height $h \in [0.25, 0.50]\,\text{mm}$. Volume $V = 100\,\text{mm}^2 \times h \approx 25\text{–}50\,\mu\text{L}$.

Capillary pressure and Lucas-Washburn fill

Capillary driving pressure across a rectangular channel ($h \ll w$):

$$\Delta P_{\text{cap}} = \gamma \cos\theta \left(\frac{1}{h} + \frac{1}{w}\right) \tag{6}$$

where γ is the sample surface tension ($\approx 72\,\text{mN/m}$ for aqueous, reduced by surfactants) and θ is the advancing contact angle (different for each surface: PDMS lid, gold floor, PETG walls).

Fill front position $L(t)$ by the Lucas-Washburn equation:

$$\frac{dL}{dt} = \frac{h^2 \Delta P_{\text{cap}}}{3\eta L} \quad \Rightarrow \quad L(t) = \sqrt{\frac{h^2 \Delta P_{\text{cap}}}{3\eta} \cdot 2t} \tag{7}$$

Fill fails if $\theta > 90°$ on any surface (reverses ΔP_{cap}).

Air venting

Without a vent, trapped air pressure rises as L increases:

$$P_{\text{air}}(L) = P_{\text{atm}} \cdot \frac{L_{\text{chamber}}}{L_{\text{chamber}} - L} \tag{8}$$

Fill stalls when $\Delta P_{\text{cap}} = P_{\text{air}}(L) - P_{\text{atm}}$. Maximum fill fraction without vent:

$$f_{\max} = 1 - \frac{P_{\text{atm}}}{\Delta P_{\text{cap}} + P_{\text{atm}}} \tag{9}$$

A vent membrane with gas permeability P_{vent} must allow air escape at the same rate as sample advances.

Electrode exposure timing

Electrode at position x_i from inlet is wetted at time t_i when $L(t_i) = x_i$. For 30-min incubation: if $t_{\text{fill}}(x_8) - t_{\text{fill}}(x_1) \ll T_{\text{incubation}}$, all electrodes see effectively equal exposure. Quantify for chosen geometry.

Layer 3: Mass transport inside the chamber (incubation)

Post-fill transport regime

After filling, the device is static. Transport is purely diffusive. Péclet number during incubation: $\text{Pe} = 0$ (no flow).

Diffusion coefficients (Stokes-Einstein, r = hydrodynamic radius):

$$D_{\text{MMP-9}} \approx \frac{k_B T}{6\pi\eta r_{\text{MMP-9}}} \approx 5 \times 10^{-11}\,\text{m}^2\text{s}^{-1} \quad (M_W = 92\,\text{kDa},\ r \approx 3.5\,\text{nm}) \tag{10}$$

$$D_{\text{HNE}} \approx 9 \times 10^{-11}\,\text{m}^2\text{s}^{-1} \quad (M_W = 30\,\text{kDa},\ r \approx 2.0\,\text{nm}) \tag{11}$$

$$D_{\text{frag}} \approx 3 \times 10^{-10}\,\text{m}^2\text{s}^{-1} \quad (\text{released MB peptide fragment},\ M_W \approx 1\,\text{kDa}) \tag{12}$$

Vertical mixing time

$t_{\text{mix}} \approx h^2/(2D)$. For $h = 0.375\,\text{mm}$, MMP-9: $t_{\text{mix}} \approx 1400\,\text{s}$. The channel is *not* well-mixed vertically during a 30-min incubation. A 1D diffusion PDE in the z-direction (perpendicular to the electrode) is required.

Coupled PDE-ODE system

Let $z \in [0, h]$ with $z = 0$ at the electrode surface.

$$\frac{\partial [E]}{\partial t} = D_E \frac{\partial^2 [E]}{\partial z^2}, \quad z \in (0, h),\ t > 0 \tag{13}$$

$$[E](z, 0) = [E]_{\text{chamber}} \quad (\text{initial condition}) \tag{14}$$

$$D_E \frac{\partial [E]}{\partial z}\bigg|_{z=0} = \frac{k_{\text{cat}} [E](0,t)\,\Gamma}{K_{m,\text{surf}} + \Gamma} \quad (\text{Robin BC, flux to surface}) \tag{15}$$

$$\frac{\partial [E]}{\partial z}\bigg|_{z=h} = 0 \quad (\text{no-flux at lid}) \tag{16}$$

Equation (15) couples the bulk PDE to the surface kinetics ODE (Layer 9).

Damköhler number

$$\mathrm{Da} = \frac{k_{\mathrm{cat}}\Gamma_0/K_{m,\mathrm{surf}}}{D_E[E]_0/h^2} \tag{17}$$

$\mathrm{Da} \ll 1$: reaction-limited (desired). $\mathrm{Da} \gg 1$: transport-limited (signal reports diffusion rate, not k_{cat}/K_m). Must verify $\mathrm{Da} < 0.1$ at expected Γ_0 and $[E]_0$.

MB fragment clearance

Released fragment ($D_{\mathrm{frag}} \approx 3 \times 10^{-10}\,\mathrm{m^2 s^{-1}}$) must diffuse away from the electrode before SWV measurement to prevent a bulk diffusion peak contaminating the surface-confined peak. Clearance time: $t_{\mathrm{clear}} \approx \delta^2/D_{\mathrm{frag}}$ where δ is the Nernst diffusion layer thickness (10–100 μm for quiescent solution). For $\delta = 50\,\mu\mathrm{m}$: $t_{\mathrm{clear}} \approx 8\,\mathrm{s}$.

Electrode cross-talk

Lateral diffusion of released MB fragment between adjacent electrodes (separation $d_{\mathrm{elec}} \sim 2\,\mathrm{mm}$): $t_{\mathrm{cross}} \approx d_{\mathrm{elec}}^2/(2D_{\mathrm{frag}}) \approx 7000\,\mathrm{s}$. Cross-talk negligible for $T_{\mathrm{incubation}} < 30\,\mathrm{min}$ provided $d_{\mathrm{elec}} > 1\,\mathrm{mm}$.

Layer 4: Gold electrode surface

Electrochemical roughness factor

True electrode area $A_{\mathrm{true}} = R_f \times A_{\mathrm{geo}}$ where $R_f \approx 1.1$–1.5 for smooth evaporated gold. Measured by integrating the gold oxide stripping charge: $R_f = Q_{\mathrm{stripping}}/(Q_{\mathrm{ref}} \times A_{\mathrm{geo}})$ where $Q_{\mathrm{ref}} = 390\,\mu\mathrm{C\,cm^{-2}}$.

All surface densities Γ are in units of $\mathrm{mol\,cm^{-2}_{true}}$.

Thin-film and contact resistance

Gold trace resistance: $R_{\mathrm{Au}} = \rho_{\mathrm{Au}} L/(wt)$ where $\rho_{\mathrm{Au}} = 2.4 \times 10^{-8}\,\Omega\,\mathrm{m}$. For $L = 5\,\mathrm{mm}$, $w = 0.5\,\mathrm{mm}$, $t = 100\,\mathrm{nm}$: $R_{\mathrm{Au}} \approx 2.4\,\Omega$. Uncompensated resistance $R_u = R_{\mathrm{Au}} + R_{\mathrm{contact}} + R_s$.

Layer 5: Reference and counter electrodes

On-chip Ag/AgCl reference

$$E_{\mathrm{ref}} = E^{\circ}_{\mathrm{AgCl/Ag}} - \frac{RT}{F}\ln(a_{\mathrm{Cl^-}}) \tag{18}$$

At $[\mathrm{Cl^-}] = 110\,\mathrm{mM}$ (wound fluid): $E_{\mathrm{ref}} \approx +0.198\,\mathrm{V}$ vs NHE $\approx -0.005\,\mathrm{V}$ vs standard Ag/AgCl.

Reference drift arises from $\mathrm{Cl^-}$ depletion at the reference junction. Acceptable drift: $< 5\,\mathrm{mV}$ over 30-min assay to avoid measurable peak position shift. Model as: $\delta E_{\mathrm{ref}}(t) = (RT/F) \times \delta \ln(a_{\mathrm{Cl}}(t))$.

Counter electrode

CE must supply current to all WEs. Max SWV peak current per WE $\approx 100\,\mathrm{nA}$; for 8 WEs: $I_{\mathrm{CE,max}} \approx 800\,\mathrm{nA}$. Require $A_{\mathrm{CE}} > 10 \times A_{\mathrm{WE}}$ (rule of thumb for non-limiting CE). Verify CE potential does not exceed O_2 evolution limit ($E > +0.97\,\mathrm{V}$ vs Ag/AgCl) during anodic SWV half-cycles.

Layer 6: Electrical double layer and solution resistance

Solution resistance

$$R_s = \frac{\rho_{\mathrm{sol}}\, d_{\mathrm{WE\text{-}RE}}}{A_{\mathrm{cell}}}, \quad \sigma_{\mathrm{wound}} \approx 15\,\mathrm{mS\,cm^{-1}} \tag{19}$$

Uncompensated iR drop during peak SWV current (100 nA): $\Delta E_{iR} = I_{\mathrm{peak}} R_s \approx 3\,\mu\mathrm{V}$ — negligible. Verify for maximum expected current.

Double-layer capacitance

For SAM-modified gold: $C_{\mathrm{dl}} \approx 1$–$5\,\mu\mathrm{F\,cm^{-2}}$ (SAM acts as thin dielectric: $C_{\mathrm{dl}} = \varepsilon_0 \varepsilon_{\mathrm{SAM}}/d_{\mathrm{SAM}}$). Capacitive background in SWV:

$$I_{\mathrm{cap}} = 2C_{\mathrm{dl}} A\, \Delta E_{\mathrm{SW}} f_{\mathrm{SWV}} \tag{20}$$

For $C_{\mathrm{dl}} = 2\,\mu\mathrm{F\,cm^{-2}}$, $A = 1\,\mathrm{mm^2}$, $\Delta E_{\mathrm{SW}} = 25\,\mathrm{mV}$, $f = 60\,\mathrm{Hz}$: $I_{\mathrm{cap}} \approx 60\,\mathrm{nA}$. Faradaic MB peak must be $> I_{\mathrm{cap}}$ to be detectable; this sets the minimum useful Γ.

Debye screening and charged residues

$\lambda_D \approx 0.8\,\mathrm{nm}$ at $I = 150\,\mathrm{mM}$. Probe tether length ($\sim 5\,\mathrm{nm}$) extends far beyond λ_D. For IPRTLTAA, the Arg residue (R, pKa 12.5) carries +1 charge at pH 7.4 and is exposed in the diffuse layer. Electrostatic repulsion from the negatively polarised electrode (at cathodic SWV potentials) may alter effective d_{eff} and hence k_{ET}. AAPVSGIG has no charged residues; IPRTLTAA must account for electrostatic correction to d_{eff}.

Layer 7: SAM formation and composition

Competitive thiol adsorption

Mixed SAM from probe-thiol (concentration c_p) and diluent-thiol (MCH or PEG-SH, concentration c_d). Equilibrium surface composition from competitive Langmuir:

$$\frac{\Gamma_p}{\Gamma_d} = \frac{c_p}{c_d} \cdot \frac{K_{a,p}}{K_{a,d}} \tag{21}$$

Short diluents adsorb preferentially ($K_{a,d} > K_{a,p}$): surface is diluent-enriched relative to solution. Solution composition must be tuned to achieve target $\Gamma_p = 1\text{–}10\,\mathrm{pmol\,cm^{-2}}$.

Phase separation

In mixed SAMs, thiols can phase-separate into nanoscale domains (5–20 nm diameter). Probe-rich domains create local steric crowding even at low mean Γ_p. Domain formation is kinetically controlled; slow deposition or annealing promotes mixing. Must assume homogeneous SAM or account for effective local crowding.

Edinburgh calcium artefact

PEG spacer chelates Ca^{2+} with association constant $K_d \approx 1\text{–}10\,\mathrm{mM}$ (literature for PEG-divalent cation complexes). At $[Ca^{2+}] = 1\,\mathrm{mM}$ in wound fluid, occupied fraction $\theta_{Ca} = [Ca^{2+}]/(K_d + [Ca^{2+}])$. Complexed PEG contracts: effective change in MB-to-Au distance:

$$\Delta d_{Ca} = \Delta d_{max} \cdot \frac{[Ca^{2+}]}{K_d + [Ca^{2+}]} \tag{22}$$

This modifies k_{ET} (Layer 10). Since the scrambled control electrode uses identical PEG architecture, the Ca^{2+} artefact is the same on active and control probes and cancels in $F_k = 1 - \rho_{active}/\rho_{control}$, *provided* Γ_p is the same on both electrodes.

Fc reference SAM

C_6-Fc-SH with diluent. Keep $\Gamma_{Fc} < 3\,\mathrm{pmol\,cm^{-2}}$ to avoid Fc-Fc through-space electron hopping (which distorts the SWV peak to an asymmetric shape). Diagnostic: symmetric SWV Fc peak confirms isolated molecules.

Layer 8: Probe architecture and conformational dynamics

Probe structure

HNE channel: Ac-Cys-[$PEG_{4/6}$]-`AAPV`↓`SGIG`-[PEG_2]-Lys(MB)-NH_2

Contour lengths: $PEG_4 = 4 \times 0.36 = 1.44\,\mathrm{nm}$; $PEG_6 = 2.16\,\mathrm{nm}$; peptide (extended) $= 8 \times 0.35 = 2.8\,\mathrm{nm}$; $PEG_2 = 0.72\,\mathrm{nm}$. Total contour $L_c \approx 5.0\text{–}6.0\,\mathrm{nm}$.

End-to-end distance distribution

Worm-like chain model (Kratky-Porod) with persistence length l_p:

$$\langle r^2 \rangle = 2 l_p L_c \left[1 - \frac{l_p}{L_c} \left(1 - e^{-L_c/l_p} \right) \right] \tag{23}$$

For PEG: $l_p \approx 0.38\,\mathrm{nm}$; for unstructured peptide: $l_p \approx 0.4\text{–}1.0\,\mathrm{nm}$. The distribution of MB-to-Au distance $P(d)$ is broad. Because $k_{ET} \propto e^{-\beta d}$, the average ET rate:

$$\langle k_{ET}^0 \rangle = \int_0^\infty k_{ET}^{max} e^{-\beta d} P(d)\, dd \tag{24}$$

is dominated by the short-d tail of $P(d)$. Must compute numerically using the WLC distance distribution.

Steric accessibility

Maximum surface coverage allowing enzyme access: $\Gamma_{steric,max} \approx 1/(\pi r_E^2)$ where r_E is the enzyme radius. For MMP-9 ($r_E \approx 2.5\,\mathrm{nm}$): $\Gamma_{steric,max} \approx 5\,\mathrm{pmol\,cm^{-2}}$. Keep $\Gamma_p < \Gamma_{steric,max}$.

Post-cleavage state

Cleavage at `AAPV`↓`SGIG`: remaining stub on surface is Ac-Cys-$PEG_{4/6}$-`AAPV` (no MB). Released fragment `SGIG`-PEG_2-Lys(MB)-NH_2 diffuses away. Only intact probes ($\Gamma_{intact} = \Gamma_0 - \Gamma_{cleaved}$) contribute to the MB SWV peak.

Layer 9: Enzyme kinetics on the surface

Surface Michaelis-Menten

$$\frac{d\Gamma}{dt} = -\frac{k_{cat}[E]_{surface}\,\Gamma}{K_{m,surf} + \Gamma} \tag{25}$$

where $[E]_{\text{surface}} = [E](z = 0, t)$ from the Layer 3 PDE, and $K_{m,\text{surf}} = K_{m,\text{sol}}/f_{\text{steric}}$ with steric correction factor $f_{\text{steric}} = \Gamma_{\text{steric,max}}/(\Gamma_{\text{steric,max}} + \Gamma_p)$ (effectively $K_{m,\text{surf}} > K_{m,\text{sol}}$ due to reduced accessibility).

First-order limit

When $\Gamma \ll K_{m,\text{surf}}$:

$$\frac{d\Gamma}{dt} \approx -\frac{k_{\text{cat}}}{K_{m,\text{surf}}}[E]_{\text{surface}}\Gamma \tag{26}$$

Solution (if $[E]_{\text{surface}} \approx$ const): $\Gamma(t) = \Gamma_0 \exp(-k_{\text{obs}}t)$ where $k_{\text{obs}} = (k_{\text{cat}}/K_{m,\text{surf}})[E]$. This is the regime where the assay measures k_{cat}/K_m.

Multi-protease competition

$$\frac{d\Gamma}{dt} = -\sum_j \frac{k_{\text{cat},j}[E_j]_{\text{surface}}\,\Gamma}{K_{m,\text{surf},j} + \Gamma} \tag{27}$$

Each interfering protease j (cathepsins, plasmin, etc.) contributes additional cleavage. Off-target contribution to F_k from protease j: $\delta F_k \approx (k_{\text{cat},j}/K_{m,\text{surf},j})[E_j]T$ at early times.

Layer 10: Marcus-theory electron transfer

Standard ET rate

$$k^0_{ET}(d) = A\exp(-\beta d)\exp\left(-\frac{\lambda}{4RT}\right) \tag{28}$$

where $\beta \approx 1.0\,\text{Å}^{-1}$ (flexible organic tether), $\lambda \approx 0.3$–$0.5\,\text{eV}$ (MB reorganisation energy), $A \approx 10^{12}$–$10^{13}\,\text{s}^{-1}$.

For $d = 3\,\text{nm}$: $k^0_{ET} \approx 10^{13} \times e^{-30} \approx 10\,\text{s}^{-1}$. Measurable in SWV at $f \sim k^0_{ET}/(2\pi) \approx 2\,\text{Hz}$; increase f to 60 Hz to operate in quasi-reversible regime for a shorter effective tether.

Butler-Volmer potential dependence

$$k_f(\eta) = k^0_{ET}\exp\left(-\frac{\alpha F\eta}{RT}\right) \quad \text{(cathodic, reduces MB)} \tag{29}$$

$$k_b(\eta) = k^0_{ET}\exp\left(\frac{(1-\alpha)F\eta}{RT}\right) \quad \text{(anodic, oxidises MB)} \tag{30}$$

where $\eta = E_{\text{applied}} - E^\circ_{\text{MB}}$ and $\alpha \approx 0.5$.

Formal potential pH correction

$E^\circ_{\text{MB}} \approx -0.25\,\text{V}$ vs Ag/AgCl at pH 7.0. pH dependence:

$$E^\circ_{\text{MB}}(\text{pH}) = E^\circ_{\text{MB}}(7.0) - 0.059(\text{pH} - 7.0)\,\text{V} \tag{31}$$

At wound pH 6.5: $E^\circ_{\text{MB}} \approx -0.22\,\text{V}$; at pH 8.0: $E^\circ_{\text{MB}} \approx -0.34\,\text{V}$. A 120 mV shift across the operating range; must include pH correction in peak fitting.

Calcium artefact correction to d_{eff}

From Layer 7: $\Delta d_{\text{Ca}} = \Delta d_{\max} \times [\text{Ca}^{2+}]/(K_d + [\text{Ca}^{2+}])$. Modified ET rate:

$$k^{0,\text{Ca}}_{ET} = k^0_{ET}(d_0) \times \exp(\beta\Delta d_{\text{Ca}}) \tag{32}$$

Cancels in ratiometric F_k provided both electrodes have identical PEG architecture and Γ_p.

Temperature correction

Through Marcus theory: $k^0_{ET} \propto T^{1/2}\exp(-\lambda/(4k_BT))$. For a 15°C temperature range: k^0_{ET} can vary by a factor of 2–3. Include temperature as explicit parameter; ratiometric design partially corrects if temperature is uniform across the chip.

Layer 11: SWV forward model

SWV waveform

Square-wave voltammetry superimposes a symmetric square-wave (amplitude ΔE_{SW}, frequency f_{SWV}) on a staircase ramp (step ΔE_{step}). Net current: $\Delta I_n = I_{\text{forward,n}} - I_{\text{backward,n}}$ at each step.

Dimensionless kinetic parameter

$$\kappa_{\mathrm{SW}} = \frac{k_{ET}^0}{f_{\mathrm{SWV}}} \cdot \frac{RT}{nF} \tag{33}$$

Regimes: $\kappa_{\mathrm{SW}} \gg 1$ (reversible, peak height $\propto \Gamma$, independent of k_{ET}^0); $\kappa_{\mathrm{SW}} \approx 1$ (quasi-reversible, optimal); $\kappa_{\mathrm{SW}} \ll 1$ (totally irreversible, peak flattens). Set f_{SWV} to achieve $\kappa_{\mathrm{SW}} \approx 1$ for the MB probe.

Numerical forward model

At each potential step n with overpotential η_n, solve Butler-Volmer for surface coverage over one half-cycle ($\Delta t = 1/(2f_{\mathrm{SWV}})$):

$$\frac{d\Gamma_{\mathrm{ox}}}{dt} = k_b(\eta_n)\Gamma_{\mathrm{red}} - k_f(\eta_n)\Gamma_{\mathrm{ox}} \tag{34}$$

$$\Gamma_{\mathrm{red}} = \Gamma_{\mathrm{intact}} - \Gamma_{\mathrm{ox}} \tag{35}$$

Net current per step: $\Delta I_n = nFA_{\mathrm{true}}(d\Gamma_{\mathrm{ox}}/dt|_{\mathrm{forward}} - d\Gamma_{\mathrm{ox}}/dt|_{\mathrm{backward}})$.

Approximate peak current

For the quasi-reversible surface case:

$$I_{\mathrm{peak,MB}} \approx \frac{n^2F^2A_{\mathrm{true}}\Gamma_{\mathrm{intact}}k_{ET}^0\Delta E_{\mathrm{SW}}}{4RT} \tag{36}$$

MB and Fc peak separation

$\Delta E^\circ = E^\circ_{\mathrm{Fc}} - E^\circ_{\mathrm{MB}} = +0.45 - (-0.25) = 0.70\,\mathrm{V}$. Peak FWHM for reversible surface wave: FWHM $= 5.56RT/(nF) \approx 65\,\mathrm{mV}$ ($n = 2$, MB) and $131\,\mathrm{mV}$ ($n = 1$, Fc). Sum of half-widths $\ll 0.7\,\mathrm{V}$: peaks are well-resolved. Verify numerically for quasi-reversible case.

Parameter optimisation

Sweep $(f_{\mathrm{SWV}}, \Delta E_{\mathrm{SW}}, \Delta E_{\mathrm{step}})$ to maximise the ratio $I_{\mathrm{peak,MB}}/I_{\mathrm{cap}}$. Recommended starting point: $f = 60\,\mathrm{Hz}$, $\Delta E_{\mathrm{SW}} = 25\,\mathrm{mV}$, $\Delta E_{\mathrm{step}} = 2\,\mathrm{mV}$. Scan window: -0.65 to $+0.65\,\mathrm{V}$ vs Ag/AgCl.

Frequency diagnostic

If measured $I_{\mathrm{peak}} \propto f^{0.5}$: diffusion-controlled (non-tethered MB in solution is dominating — wrong regime). If $I_{\mathrm{peak}} \propto f^1$ (at low f) or plateaus (at high f): surface-confined, correct. Must verify.

Layer 12: SWV peak extraction and signal processing

Raw data processing

Typical scan: $(\Delta E_{\mathrm{total}}/\Delta E_{\mathrm{step}}) \approx 650$ points. Steps:

1. Savitzky-Golay smoothing (window = 11 pts, 3rd-order polynomial) to reduce high-frequency noise while preserving peak shape.
2. Cubic spline baseline through non-peak windows (e.g., -0.65 to $-0.45\,\mathrm{V}$ and -0.05 to $+0.25\,\mathrm{V}$ for MB channel).
3. Gaussian peak fit: $\Delta I(E) = A_{\mathrm{peak}} \exp[-(E - E_p)^2/(2\sigma_E^2)]$ with parameters $(A_{\mathrm{peak}}, E_p, \sigma_E)$ fitted by nonlinear least squares.
4. Peak area: $\mathcal{A} = A_{\mathrm{peak}}\sigma_E\sqrt{2\pi}$ (integrated area is more robust to k_{ET}-frequency mismatch than peak height).

Quality diagnostics per scan

- Peak width $\sigma_E > 2\times$ expected ($> 65\,\mathrm{mV}$ for MB): SAM heterogeneous or measurement error; flag.
- Peak position shift $|\delta E_p| > 5\,\mathrm{mV}$ from scan to scan: reference drift or SAM degradation; flag.
- Fit residuals $> 3\sigma$: two-population MB (heterogeneous tether lengths) or non-Gaussian artifact; fit with two-Gaussian model.

Layer 13: Ratiometric signal and error propagation

Signal definitions

For electrode i at time T, using peak areas $\mathcal{A}_{\mathrm{MB}}$ and $\mathcal{A}_{\mathrm{Fc}}$:

$$R_i(T) = \frac{\mathcal{A}_{\mathrm{MB},i}(T)}{\mathcal{A}_{\mathrm{Fc},i}(T)}, \qquad \rho_i = \frac{R_i(T)}{R_i(0)}, \qquad F_k = 1 - \frac{\rho_{\mathrm{active,k}}}{\rho_{\mathrm{control,k}}} \tag{37}$$

$$\kappa_k = -\frac{\ln(1 - F_k)}{T} \tag{38}$$

Error propagation

$$\frac{\sigma_R}{R} = \sqrt{\left(\frac{\sigma_{\mathrm{MB}}}{\mathcal{A}_{\mathrm{MB}}}\right)^2 + \left(\frac{\sigma_{\mathrm{Fc}}}{\mathcal{A}_{\mathrm{Fc}}}\right)^2} \tag{39}$$

$$\frac{\sigma_\rho}{\rho} = \sqrt{2} \cdot \frac{\sigma_R}{R} \quad \text{(equal variance for } R(0) \text{ and } R(T)\text{)} \tag{40}$$

$$\sigma_F = \frac{\rho_a}{\rho_c}\sqrt{\left(\frac{\sigma_{\rho_a}}{\rho_a}\right)^2 + \left(\frac{\sigma_{\rho_c}}{\rho_c}\right)^2} \tag{41}$$

$$\sigma_\kappa = \frac{\sigma_F}{(1-F_k)T} \tag{42}$$

Note: σ_κ diverges as $F_k \to 1$, setting the upper dynamic range ceiling.

Dynamic range

Useful range: $F_k \in [3\sigma_F,\ 0.90]$, corresponding to:

$$\kappa_k \in \left[\frac{3\sigma_F}{T},\ \frac{2.3}{T}\right] \tag{43}$$

Fouling cancellation and its limits

The double-ratio F_k cancels fouling that affects $\mathcal{A}_{\mathrm{MB}}$ equally on active and control probes. This holds if: (a) fouling is a bulk matrix effect (protein adsorption); (b) both electrodes have identical SAM diluent composition. If fouling is sequence-specific (albumin preferentially adsorbs to one peptide sequence), cancellation is incomplete. This assumption must be tested in the calibration experiment.

Layer 14: Electronics and potentiostat noise budget

TIA gain selection

Max expected current (all WEs, including capacitive): $I_{\max} \approx 200\,\mathrm{nA}$ per WE. For output voltage $< 1\,\mathrm{V}$: $R_f < 5\,\mathrm{M\Omega}$. Choose $R_f = 100\,\mathrm{k\Omega}$ (safety factor 10; noise scales as $1/\sqrt{R_f}$).

Noise sources

$$i_{\mathrm{Johnson}} = \sqrt{\frac{4k_BT \cdot \mathrm{BW}}{R_f}} \approx 4\,\mathrm{pA} \quad (R_f = 100\,\mathrm{k\Omega},\ \mathrm{BW} = 100\,\mathrm{Hz}) \tag{44}$$

$$i_{\mathrm{shot}} = \sqrt{2eI_{\mathrm{DC}}\,\mathrm{BW}} \approx 44\,\mathrm{fA} \quad (I_{\mathrm{DC}} = 60\,\mathrm{nA}) \tag{45}$$

$$i_{\mathrm{ADC}} = \frac{I_{\mathrm{range}}}{2^N\sqrt{12}} \approx 2\,\mathrm{fA} \quad (N = 24\,\mathrm{bit},\ I_{\mathrm{range}} = 100\,\mathrm{nA}) \tag{46}$$

Total: $i_{\mathrm{total}} \approx i_{\mathrm{Johnson}} \approx 4\,\mathrm{pA/Hz^{1/2}}$. Dominant noise source is Johnson noise in R_f.

Multiplexer settling

After switching WE: settling time $\tau = (R_{\mathrm{mux}} + R_s)C_{\mathrm{dl}}A \approx 33\,\Omega \times 2\,\mathrm{nF} = 66\,\mathrm{ns}$. Negligible vs SWV half-period of 8 ms.

Mains interference

50 Hz (EU mains) is close to SWV frequency $f = 60\,\mathrm{Hz}$. Beating frequency $|60 - 50| = 10\,\mathrm{Hz}$ creates slow modulation of the SWV baseline. Mitigation: shielded enclosure; or set $f_{\mathrm{SWV}} = 50\,\mathrm{Hz}$ synchronised to mains; or $f_{\mathrm{SWV}} \gg 100\,\mathrm{Hz}$ with appropriate κ_{SW} adjustment.

Required SNR

To resolve $F_{\mathrm{LOD}} = 5\%$: require $\sigma_F < 1.7\%$. With $\mathcal{A}_{\mathrm{MB}} \approx 1\,\mathrm{pA \cdot V}$ and $\sigma_{\mathcal{A}} \approx 0.4\,\mathrm{fA \cdot V}$ (from i_{Johnson} integrated over ~ 50 SWV points): $\sigma_{\mathcal{A}}/\mathcal{A} \approx 0.04\%$. Full propagation through $R \to \rho \to F$ gives $\sigma_F \approx 0.06\%$ — well below the 1.7% target.

Layer 15: Calibration curve and limit of detection

Calibration model

In the first-order regime ($F_k \ll 1$):

$$F_k \approx \frac{k_{\mathrm{cat}}}{K_{m,\mathrm{surf}}}[E]_{\mathrm{chamber}}\,T \tag{47}$$

Linear in $[E]$; slope = $(k_{\text{cat}}/K_{m,\text{surf}}) \times T$ (from pES-01 to pES-08 wet lab + steric correction).

LOD and LLOQ

$$\text{LOD} = \frac{3\sigma_{\text{blank}}}{dF_k/d[E]}, \qquad \text{LLOQ} = \frac{10\sigma_{\text{blank}}}{dF_k/d[E]} \tag{48}$$

Expected: with $\sigma_F \approx 1\%$ and slope $\approx (10^4\,\text{M}^{-1}\text{s}^{-1}) \times 1800\,\text{s} = 1.8 \times 10^7\,\text{M}^{-1}$:

$$\text{LOD} \approx \frac{0.03}{1.8 \times 10^7} \approx 1.7\,\text{nM} \tag{49}$$

Clinical MMP-9 range in wound fluid: 1–100 nM. Target LOD < 1 nM after optimisation.

Two-channel deconvolution

Observed signal vector: $\mathbf{F} = [F_{\text{MMP9}}, F_{\text{HNE}}]^\top$. Sensitivity matrix:

$$\mathbf{F} = \mathbf{S} \cdot \mathbf{c}, \qquad \mathbf{S} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \tag{50}$$

where $S_{ij} = dF_{\text{ch},i}/d[E_j]$ (on-diagonal: specific; off-diagonal: cross-sensitivity). Estimated concentrations:

$$\hat{\mathbf{c}} = \mathbf{S}^{-1}\mathbf{F}, \qquad \sigma_{\hat{c}} = \mathbf{S}^{-1}\sigma_{\mathbf{F}} \tag{51}$$

Condition number $\kappa(\mathbf{S}) = \|\mathbf{S}\| \cdot \|\mathbf{S}^{-1}\|$ quantifies deconvolution noise amplification. Target: $\kappa(\mathbf{S}) < 30$ (same criterion as Panel A/B substrate selectivity). This directly links the substrate design dry-lab work to the device model: IPRTLTAA and AAPVSGIG were selected to minimise off-diagonal elements of $\mathbf{S}$, keeping $\kappa(\mathbf{S})$ low.

Layer 16: Full chain integration and falsifiable predictions

Forward simulation chain

Given inputs ($[E]_{\text{MMP-9,wound}}$, $[E]_{\text{HNE,wound}}$, T, pH, $[\text{Ca}^{2+}]$, T_{amb}):

1. Layer 0: ODE → active fractions at time of sampling
2. Layers 1–2: dilution chain → $[E]_{\text{chamber}}$
3. Layer 3: PDE (discretised, $N_z = 50$ grid points) → $[E](z = 0, t)$
4. Layer 7: competitive Langmuir → Γ_p, Γ_d
5. Layer 8: WLC model → $P(d)$ → $\langle k^0_{ET} \rangle$; steric factor
6. Layer 9: surface MM ODE (coupled to PDE) → $\Gamma_{\text{intact}}(t)$
7. Layer 10: Marcus, pH, Ca^{2+}, temperature corrections → $k^0_{ET,\text{eff}}$
8. Layer 11: SWV numerical model → $\mathcal{A}_{\text{MB}}(T)$, $\mathcal{A}_{\text{Fc}}(T)$
9. Layers 12–13: peak extraction, R, ρ, F_k, κ_k
10. Layer 14: add instrument noise
11. Layer 15: calibration lookup → $[\hat{E}]_{\text{MMP-9}}, [\hat{E}]_{\text{HNE}}$

Falsifiable outputs

- Predicted $F_k(T)$ vs measured $F_k(T)$ from calibration experiment in wound-fluid matrix.
- Predicted LOD vs measured LOD.
- Predicted $\kappa(\mathbf{S})$ vs measured cross-sensitivity (iGEM gold criterion: model predicts, wet lab confirms).
- Predicted Damköhler number regime vs observed F_k frequency dependence on $[E]$.

Global sensitivity analysis

Sobol variance decomposition (SALib) over all uncertain parameters to identify which layer contributes most variance to σ_{F_k}. Expected dominant parameters: Γ_p (SAM), $k_{\text{cat}}/K_{m,\text{surf}}$ (wet lab), d_{eff} (conformational model). Results will direct where to invest experimental effort to tighten the model.

Negative control: scrambled AAPVSGIG

Charge balance

AAPVSGIG composition: A×2, P×1, V×1, S×1, G×2, I×1. All residues are neutral at physiological pH 7.4 (no charged side chains). The probe termini are Ac (N) and NH_2 (C), both uncharged. Net charge = 0 for any permutation of these 8 residues. "Same charge balance" is automatically satisfied by any scramble preserving amino acid composition.

Scramble selection protocol

1. Enumerate unique permutations of {A, A, P, V, S, G, I, G}: total $8!/(2! \times 2!) = 10{,}080$.
2. Filter: remove sequences containing the `AAPV` motif (HNE P4-P1 recognition) at any position.
3. Score each remaining candidate with the existing HNE logistic regression + PSSM scorer and with the MMP-9 model.
4. Select the candidate with the lowest p_{HNE} (LR probability). Break ties by lowest $p_{\text{MMP-9}}$.
5. Report: selected sequence, p_{HNE}, PSSM_{HNE}, $p_{\text{MMP-9}}$, $\text{PSSM}_{\text{MMP-9}}$.

Selected scramble is used on all control electrodes (paired with each active channel electrode) throughout the cartridge.

Literature parameters required

Parameter	Symbol	Value	Source
MMP-9 diffusion coefficient	$D_{\text{MMP-9}}$	$\sim 5 \times 10^{-11}\,\text{m}^2\text{s}^{-1}$	Stokes-Einstein / experimental
HNE diffusion coefficient	D_{HNE}	$\sim 9 \times 10^{-11}\,\text{m}^2\text{s}^{-1}$	Stokes-Einstein / experimental
Tunnelling decay constant	β	$\sim 1.0\,\text{Å}^{-1}$	Smalley et al. 2003
MB reorganisation energy	λ	0.3–0.5 eV	Pheeney and Barton 2012
MB formal potential	E°_{MB}	−0.25 V (pH 7)	Kelley et al. 1997
Fc formal potential	E°_{Fc}	+0.45 V	Nowicka et al.
SAM double-layer capacitance	C_{dl}	1–5 $\mu\text{F cm}^{-2}$	Campuzano et al.
Optimal probe surface coverage	Γ_p	1–10 pmol cm^{-2}	Lubin and Plaxco 2010
PEG persistence length	l_p	~ 0.38 nm	Oesterhelt et al. 1999
Gold roughness factor	R_f	1.1–1.5	Electrochemical literature
Wound fluid $[Ca^{2+}]$	—	1–2 mM	Gethin et al.
Wound fluid pH range	—	6.5–8.2	Gethin et al.
k_{cat}/K_m MMP-9 on IPRTLTAA	—	from pES	pES-01 to pES-08 wet lab
k_{cat}/K_m HNE on AAPVSGIG	—	from pES	pES-01 to pES-08 wet lab

Implementation plan

1. **Immediate**: generate scrambled AAPVSGIG negative control using `substrate_predictor/`.
2. **Layer 3 + 9**: implement coupled PDE-ODE in `models/03_fluidics/` and `models/05_surface_kinetics/` (Python, `scipy.integrate`). These are the critical kinetic layers.
3. **Layer 11**: implement SWV numerical forward model in `models/07_swv_forward/`. This is the single most important modelling component.
4. **Layer 8**: implement WLC distance distribution and compute $\langle k^0_{ET} \rangle$ in `models/06_electron_transfer/`.
5. **Layers 7, 14, 15**: SAM composition model, noise budget, calibration in their respective folders.
6. **Layer 16**: chain all layers into a single forward simulation in `models/16_forward_chain/forward_chain.py`.
7. **Sensitivity analysis**: SALib Sobol in `models/16_forward_chain/sensitivity.py`.
8. Populate `data/literature_parameters/` with structured CSV of all numeric parameters and DOI sources before implementing any layer.